

The reef that came back after 150 years

30% more reef

Researchers watched the seabed under the UK's first large offshore mussel farm, in Lyme Bay, for **eight years**. Compared with control sites nearby, the damaged seabed under the farm got more complex, and **biogenic reef increased by 30% after seven years**. Shellfish reef is returning to that part of the seabed for the first time in up to 150 years.

• WHAT TO DRAW OUT

Mussels are not only food. Shells that drop off the ropes build hard ground on bare sand, and that new ground is what a lot of the animals on the mat are living on. Ask which cards people moved onto the structure.

Mascorda-Cabre, L., Hosegood, P., Attrill, M.J. & Sheehan, E.V. (2024) Biogenic reef creation and biodiversity enhancement by an offshore longline mussel farm. *Ecological Indicators*, 167: 112708. doi:10.1016/j.ecolind.2024.112708

· CUT HERE ·

More fish, and young ones settling in

1.6× more fish

Pulling together studies from around the world, fish were on average **1.6 times more abundant on shellfish and seaweed farms** than in comparable habitat nearby. In one study, the larvae of **seven different fish species** settled out of the plankton into farm structures, at a diversity similar to natural habitat.

• WHAT TO DRAW OUT

The difference between attracting and producing. Adult fish moving in from nearby is not the same as young fish settling and growing up there, and only the second one earns the word "nursery". Ask the room which of the two their mat actually shows.

Theuerkauf, S.J., Barrett, L.T., Alleway, H.K., Costa-Pierce, B.A., St. Gelais, A. & Jones, R.C. (2021) Habitat value of bivalve shellfish and seaweed aquaculture for fish and invertebrates: pathways, synthesis and next steps. *Reviews in Aquaculture*, 14(1): 54-72. doi:10.1111/raq.12584

Crab and lobster do well out of it

Commercial species gain

The same Lyme Bay farm was studied specifically for its effect on species people actually catch. The work found a **restorative function for commercial crustaceans**, with the farm structure and the material falling from it supporting more crab and lobster than the surrounding seabed. The operators report large increases in bass, mullet, crab and lobster over nine years.

• WHAT TO DRAW OUT

Follow the arrows upward. Hermit crabs and shore crabs eat what rains down off the ropes, and the things that eat *them*, including species that end up in a pot, gain too. This is the card that connects the mat to somebody's livelihood.

Stamp, T., Pittman, S.J., Holmes, L.A., Rees, A., Ciotti, B.J., Thatcher, H., Davies, P., Hall, A., Wells, G., Olczak, A. & Sheehan, E.V. (2024) Restorative function of offshore longline mussel farms with ecological benefits for commercial crustacean species. *Science of the Total Environment*, 951: 174987. doi:10.1016/j.scitotenv.2024.174987

· CUT HERE ·

A patch of easier water for shellfish

23% easier shell-building

Seaweed photosynthesises, which strips carbon dioxide out of the water around it. Measured inside a kelp farm at peak growth, the water was **less acidic than outside**, and aragonite saturation, the measure of how easily an animal can build a shell, was **23% higher**. Seaweed farms have been described as local refuges from ocean acidification.

• WHAT TO DRAW OUT

Limpets, urchins and top shells all build shells out of seawater, and as the sea slowly acidifies that gets harder. Inside the canopy it is measurably easier. Note the limits out loud: the effect is local, and strongest at peak growth.

Xiao, X., Agustí, S., Yu, Y., Huang, Y., Chen, W., Hu, J., Li, C., Li, K., Wei, F., Lu, Y., Xu, C., Chen, Z., Liu, S., Zeng, J., Wu, J. & Duarte, C.M. (2021) Seaweed farms provide refugia from ocean acidification. *Science of the Total Environment*, 776: 145192. doi:10.1016/j.scitotenv.2021.145192

The mud underneath gets richer, and not everything likes that

7 m of altered seabed

Mussels drop faeces and pseudofaeces onto the seabed all year. On the west coast of Scotland, sediment next to the farm lines was measurably **less oxygenated**, with redox potential down by about **50 millivolts**, an effect still detectable **up to about seven metres** from the farm before it faded away. How bad this gets depends almost entirely on what was there first. On already-enriched mud it can tip into airless sediment; on clean, well-flushed sand it can simply mean more food.

"What does enriched sediment mean, and why might that push the clean-sand species out?"

• WHAT TO DRAW OUT

The sea potato is the card to reach for. It is a burrower that wants clean, well-sorted sand, and organic sludge is the wrong habitat for it. A farm does not improve a place for everything living there. It changes it, and some animals were suited to what it was before.

Wilding, T.A. (2012) Changes in sedimentary redox associated with mussel (*Mytilus edulis* L.) farms on the west-coast of Scotland. *PLoS ONE*, 7(9): e45159. doi:10.1371/journal.pone.0045159. See also Wilding, T.A. & Nickell, T.D. (2013) *PLoS ONE*, 8(7): e68313. doi:10.1371/journal.pone.0068313

- CUT HERE -

The habitat is a crop, and crops get taken away

Seasonal not permanent

A reef stays. A farm is grown, harvested and regrown. The kelp that shelters a shoal and the mussel clumps that a whelk lives on are **removed on a schedule**, and everything living on them goes with them or has to move. Reviews of aquaculture habitat value make the point directly: the benefit depends on the structure being there, so how often the crop comes out, how much is left behind, and whether anything is left standing over winter all change the answer.

"What happens to everything you have just drawn when the crop is harvested?"

• WHAT TO DRAW OUT

This is the single best question for showing that timing matters. A survey in August and a survey in February at the same farm can tell you opposite things. Ask what a farmer could change about harvesting to keep more of the web intact.

Theuerkauf, S.J., Barrett, L.T., Alleway, H.K., Costa-Pierce, B.A., St. Gelais, A. & Jones, R.C. (2021) Habitat value of bivalve shellfish and seaweed aquaculture for fish and invertebrates: pathways, synthesis and next steps. *Reviews in Aquaculture*, 14(1): 54-72. doi:10.1111/raq.12584

The anchors have a footprint of their own

Scour

around every mooring

A farm is held down by anchors and chains, and those chains move. As the lines swing with wind and tide the chain sweeps the seabed around each mooring, shifting sediment and scraping off whatever is growing there. A review of the evidence across five seabed types found **every study reported a negative response from the animals and plants living there**. On bare sand that footprint is small. Dropped on a living reef or a seagrass bed, it is not.

"Can the anchors damage natural reef, and does it matter where you put the farm?"

• WHAT TO DRAW OUT

Siting is the whole argument. The case for a farm building habitat is strongest on bare, already-damaged ground and weakest on top of something that was doing the job already. This is also the honest answer to anyone who suspects we are only here to sell farms.

Broad, A., Rees, M.J. & Davis, A.R. (2020) Anchor and chain scour as disturbance agents in benthic environments: trends in the literature and charting a course to more sustainable boating and shipping. *Marine Pollution Bulletin*, 161: 111683. doi:10.1016/j.marpolbul.2020.111683

· CUT HERE ·

A farm is not a wild kelp forest

Fewer

species than the real thing

A 2025 study of wave-exposed kelp farms in Norway used both eDNA and traditional sampling, netting, cameras and invertebrate collection. Compared with **natural habitats nearby**, the kelp farms held **lower species richness and lower abundance**. A farm adds habitat to bare ground. It does not replace a real reef, and it should never be an argument for damaging one.

"Are there species that simply do not want to live on the infrastructure?"

• WHAT TO DRAW OUT

The honest ceiling on the whole story. Farms help most where there was nothing to begin with. They are not a substitute for protecting what is already there, and anyone who tells you otherwise is selling something.

Andrews, A.J., Christie, H., Praebel, K., Brandner, M., Torstensen, R.G., Lewis, L., Hestness, R.B., Berg, P.R., Bekkby, T. & Hancke, K. (2025) Combining environmental DNA and traditional sampling to assess the role of kelp aquaculture as an artificial habitat in Norway. *Frontiers in Marine Science*, 12: 1717725. doi:10.3389/fmars.2025.1717725

The carbon claim does not stand up

You will hear that seaweed farming locks away large amounts of carbon. Recent assessments do not support it. Globally, seaweed farming is currently estimated to be a **net emitter**: the carbon sequestered is more than cancelled out by the emissions of running the farms. There is **no standardised way to verify** seaweed carbon storage, unlike saltmarsh or seagrass. Where seaweed genuinely helps the climate is by **replacing something worse**, such as fertiliser, animal feed or plastic, not by burying carbon.

• WHY WE SAY THIS OUT LOUD

Somebody in the room already believes the carbon story. Correcting it yourself, before you are challenged, is the fastest way to be trusted on everything else you say.

Pessarrodona, A., Howard, J., Pidgeon, E., Wernberg, T. & Filbee-Dexter, K. (2024) Carbon removal and climate change mitigation by seaweed farming: a state of knowledge review. *Science of the Total Environment*, 918: 170525. doi:10.1016/j.scitotenv.2024.170525

- CUT HERE -

THE GAP WE ARE FILLING

PEBL

Almost none of this was measured in Wales

Nearly every number on these cards comes from Lyme Bay, Norway, Scotland or China. The evidence that farms build habitat is real, but it is **somewhere else**, on other seabeds, with other species, at other scales. Welsh farms are smaller, closer inshore and mostly newer.

That is why PEBL puts cameras on them. Everything built on the mat today came out of footage from real farms, identified frame by frame. **The clips playing at the start are the raw evidence.**

• WHAT YOU CAN DO

The code on your species card opens that footage. You can help identify what our cameras are seeing, from home, in ten minutes. That is not a gimmick, the identifications feed the dataset.

Where these numbers come from

1. Mascorda-Cabre, L., Hosegood, P., Attrill, M.J. & Sheehan, E.V. (2024) Biogenic reef creation and biodiversity enhancement by an offshore longline mussel farm. *Ecological Indicators*, 167: 112708. doi:10.1016/j.ecolind.2024.112708
2. Stamp, T., Pittman, S.J., Holmes, L.A., Rees, A., Ciotti, B.J., Thatcher, H., Davies, P., Hall, A., Wells, G., Olczak, A. & Sheehan, E.V. (2024) Restorative function of offshore longline mussel farms with ecological benefits for commercial crustacean species. *Science of the Total Environment*, 951: 174987. doi:10.1016/j.scitotenv.2024.174987
3. Theuerkauf, S.J., Barrett, L.T., Alleway, H.K., Costa-Pierce, B.A., St. Gelais, A. & Jones, R.C. (2021) Habitat value of bivalve shellfish and seaweed aquaculture for fish and invertebrates: pathways, synthesis and next steps. *Reviews in Aquaculture*, 14(1): 54-72. doi:10.1111/raq.12584
4. Xiao, X., Agusti, S., Yu, Y., Huang, Y., Chen, W., Hu, J., Li, C., Li, K., Wei, F., Lu, Y., Xu, C., Chen, Z., Liu, S., Zeng, J., Wu, J. & Duarte, C.M. (2021) Seaweed farms provide refugia from ocean acidification. *Science of the Total Environment*, 776: 145192. doi:10.1016/j.scitotenv.2021.145192
5. Wilding, T.A. (2012) Changes in sedimentary redox associated with mussel (*Mytilus edulis* L.) farms on the west-coast of Scotland. *PLoS ONE*, 7(9): e45159. doi:10.1371/journal.pone.0045159
6. Wilding, T.A. & Nickell, T.D. (2013) Changes in benthos associated with mussel (*Mytilus edulis* L.) farms on the west-coast of Scotland. *PLoS ONE*, 8(7): e68313. doi:10.1371/journal.pone.0068313

Findings are summarised in plain English for a public audience. Figures are quoted as reported by the authors. Where a study disagrees with the others we have included it rather than left it out.

- CUT HERE -

Where these numbers come from

7. Broad, A., Rees, M.J. & Davis, A.R. (2020) Anchor and chain scour as disturbance agents in benthic environments: trends in the literature and charting a course to more sustainable boating and shipping. *Marine Pollution Bulletin*, 161: 111683. doi:10.1016/j.marpolbul.2020.111683
8. Andrews, A.J., Christie, H., Praebel, K., Brandner, M., Torstensen, R.G., Lewis, L., Hestness, R.B., Berg, P.R., Bekkby, T. & Hancke, K. (2025) Combining environmental DNA and traditional sampling to assess the role of kelp aquaculture as an artificial habitat in Norway. *Frontiers in Marine Science*, 12: 1717725. doi:10.3389/fmars.2025.1717725
9. Pessarrodona, A., Howard, J., Pidgeon, E., Wernberg, T. & Filbee-Dexter, K. (2024) Carbon removal and climate change mitigation by seaweed farming: a state of knowledge review. *Science of the Total Environment*, 918: 170525. doi:10.1016/j.scitotenv.2024.170525
10. Mascorda Cabre, L., Hosegood, P., Attrill, M.J., Bridger, D. & Sheehan, E.V. (2021) Offshore longline mussel farms: a review of oceanographic and ecological interactions to inform future research needs, policy and management. *Reviews in Aquaculture*, 13(4): 1864-1887. doi:10.1111/raq.12549

Species biology on the cards themselves is drawn from the Marine Life Information Network (Tyler-Walters, H., ed. *MarLIN: Biology and Sensitivity Key Information Reviews*, Marine Biological Association of the UK), Hayward, P.J. & Ryland, J.S. (2017) *Handbook of the Marine Fauna of North-West Europe*, 2nd edn, Oxford University Press, and FishBase (Froese, R. & Pauly, D., eds).